

AnomX

Trader Behavior Anomaly Detection Engine

Mentored by **Sugandh Kumar**

A beginner-friendly yet industry-oriented Summer of Code project focused on building a real-time anomaly detection system for suspicious trading and user behavior using Machine Learning, Streaming Systems, and APIs.

Seasons of Code 2026
Indian Institute of Technology Bombay

Project Introduction

Modern financial platforms continuously process massive streams of user activity such as logins, deposits, withdrawals, and trading events. Detecting suspicious or unusual behavior in such systems is extremely important for fraud prevention and risk management.

In this project, we will build an end-to-end AI-powered anomaly detection system capable of identifying suspicious trader behavior in near real-time.

What We Will Build

By the end of the project, mentees will collaboratively build:

- A synthetic financial activity dataset
- A feature engineering pipeline
- Baseline anomaly detection models
- Deep learning based sequence models
- Real-time streaming simulation
- A FastAPI inference server
- Explainable anomaly alerts

The final system will simulate how real financial platforms monitor suspicious activities like:

- Unusual login behavior
- Rapid IP switching
- Sudden withdrawal spikes

- Abnormal trading activity
- Burst behavioral patterns

These ideas are inspired by real anomaly detection requirements commonly seen in trading and fraud monitoring systems.

Who Is This Project For?

This project is intentionally designed to be beginner-friendly.

You **do not** need prior experience in:

- Machine Learning
- Deep Learning
- Kafka or Streaming Systems
- APIs or Deployment

We will cover the required fundamentals throughout the mentorship period.

However, mentees should ideally:

- Be comfortable with basic programming
- Be willing to learn Python
- Have curiosity towards AI/ML systems
- Be interested in building real-world systems

Technologies We Will Explore

Area	Tools / Concepts
Programming	Python, Git, VS Code
Data Handling	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
Deep Learning	PyTorch
Streaming	Kafka / Redpanda
Backend APIs	FastAPI
Deployment	Docker, Render (Optional)

10-Week Roadmap

Weeks 1–2 : Foundations

- Python basics and project setup
- Git and GitHub workflow
- Introduction to anomaly detection
- Understanding the project architecture
- Data handling using Pandas

Weeks 3–4 : Data and Features

- Synthetic dataset generation
- Exploratory Data Analysis (EDA)
- Feature engineering basics
- Behavioral and temporal metrics

Weeks 5–6 : Classical Machine Learning

- Isolation Forest / LOF
- Model evaluation
- Understanding anomaly scores
- Visualizing anomalies

Weeks 7–8 : Deep Learning for Sequences

- Sequential data understanding
- LSTM basics
- Autoencoders
- Real-time behavioral modeling

Week 9 : Streaming Systems

- Introduction to Kafka / Redpanda
- Producer-consumer architecture
- Simulating event streams

Week 10 : APIs and Final Integration

- FastAPI basics
- Serving ML models
- Explainable alerts
- Final project integration
- Demo and presentations

What You Will Learn

By the end of the mentorship, mentees will gain exposure to:

- Real-world anomaly detection systems
- Machine learning workflows
- Sequence modeling using LSTMs
- Feature engineering techniques
- Streaming data pipelines
- API development
- End-to-end system integration

More importantly, mentees will understand how modern AI systems are designed beyond just training models.

Final Note

This project is not only about Machine Learning.

It is about learning how different parts of a real-world AI system connect together — data engineering, feature pipelines, model training, streaming systems, APIs, and explainability.

The goal is to create an environment where beginners can comfortably explore advanced ideas while collaboratively building something meaningful and practical.

Let's build AnomX together!